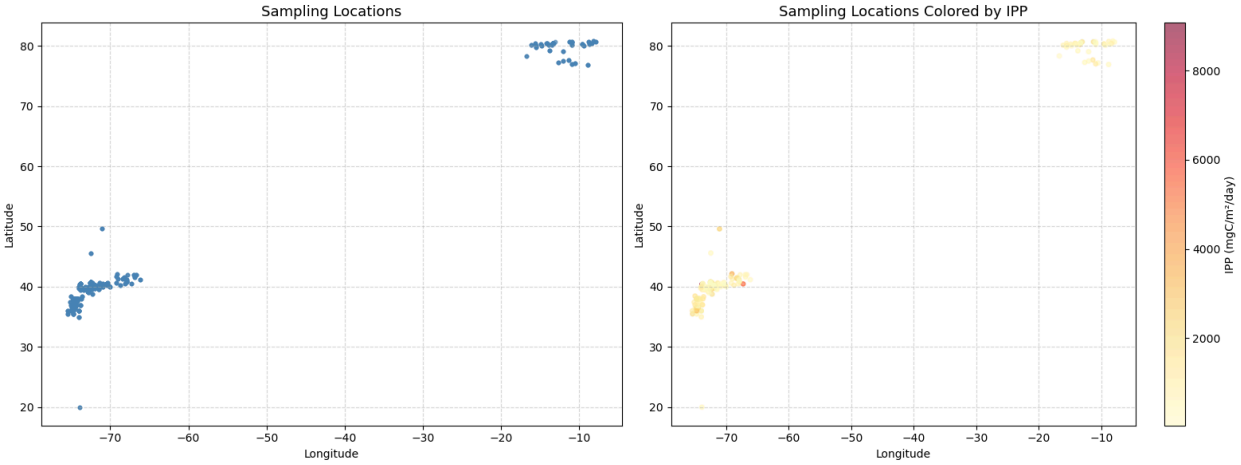
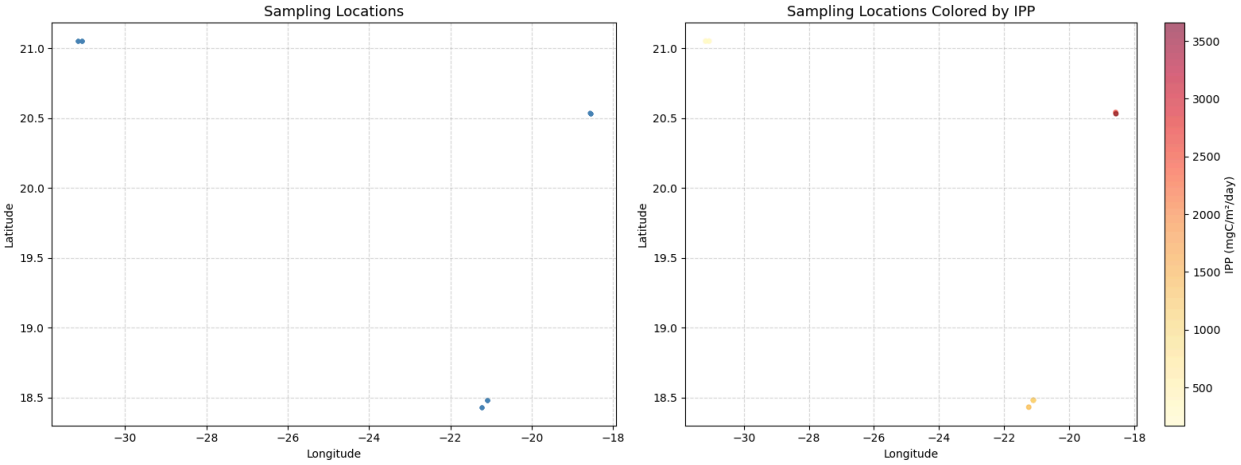
Overview

1. About the dataset:
    a. Eight datasets were loaded from the Behrenfeld-Falkowski ocean primary productivity database. Each file contains oceanographic measurements collected by research vessels across multiple ocean regions and time periods. The data captures depth-resolved measurements of chlorophyll concentration, photosynthesis rates, and integrated primary production (IPP) at various geographic locations.
2. Structure of the files:
    a. All files share the same 18-column structure:
        i. Count
        ii. Cruise
        iii. Day
        iv. Month
        v. Year
        vi. LAT
        vii. LONG
        viii. Temp
        ix. DayLength
        x. Irr
        xi. Pb_opt
        xii. Depth
        xiii. %Irr
        xiv. O.D.
        xv. Chl
        xvi. Pb
        xvii. PP
        xviii. IPP
3. We have a total of 20558 rows in all the files

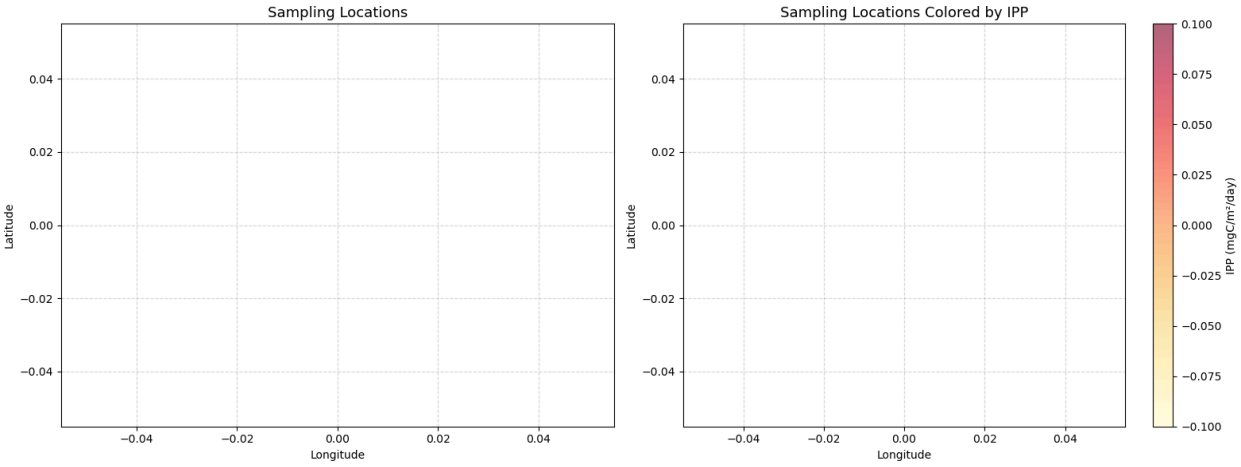
4. Key observations across the different datasets:
    a. Dataset 1:
        i. US East Coast + Arctic: Two distinct geographic clusters are visible. One centered around LAT 35 – 45°N, LONG -70 to -75° (US East Coast shelf waters), and a second cluster at LAT 77– 80°N (Arctic Ocean). This file has the broadest latitudinal range of all datasets, spanning over 60 degrees of latitude.



    b. Dataset 2:
        i. Tropical Atlantic (Eumeli): Only 4–5 unique sampling stations are present, located between LAT 18–21°N, LONG -18 to -31°. This is a sparse dataset geographically. The IPP-colored plot reveals one notably high-productivity station near the African coast (LONG ~-19), consistent with coastal upwelling dynamics in that region, while open-ocean stations showed lower productivity.

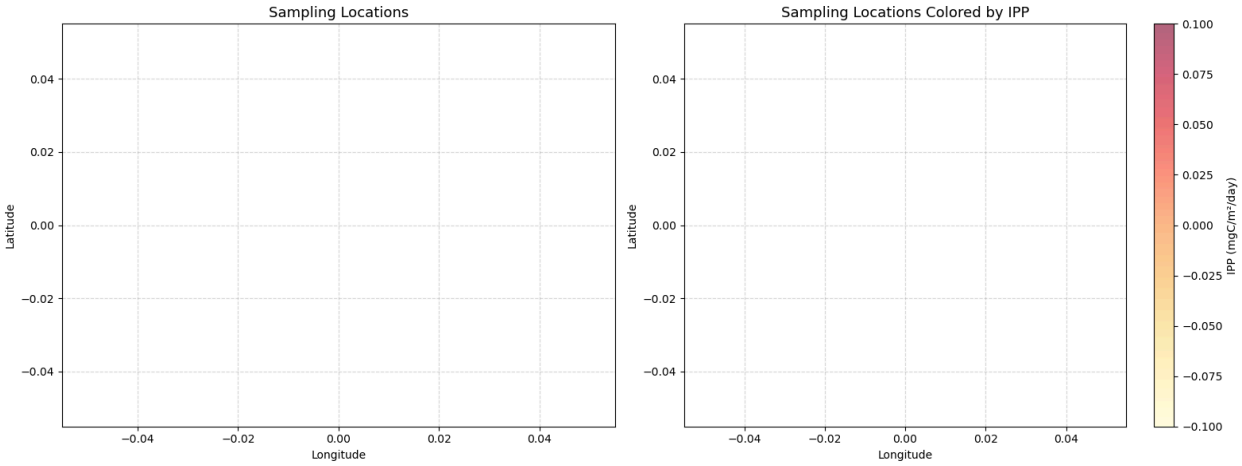


c. Dataset 3:
    i. HOTS (Hawaii): Longitude data is missing for this dataset (NaN in the LONG column), so the geographic plot is limited. Sampling is concentrated around LAT ~21°N, consistent with the Hawaii Ocean Time-series station location in the North Pacific subtropical gyre. This is a characteristically low-productivity oligotrophic region.

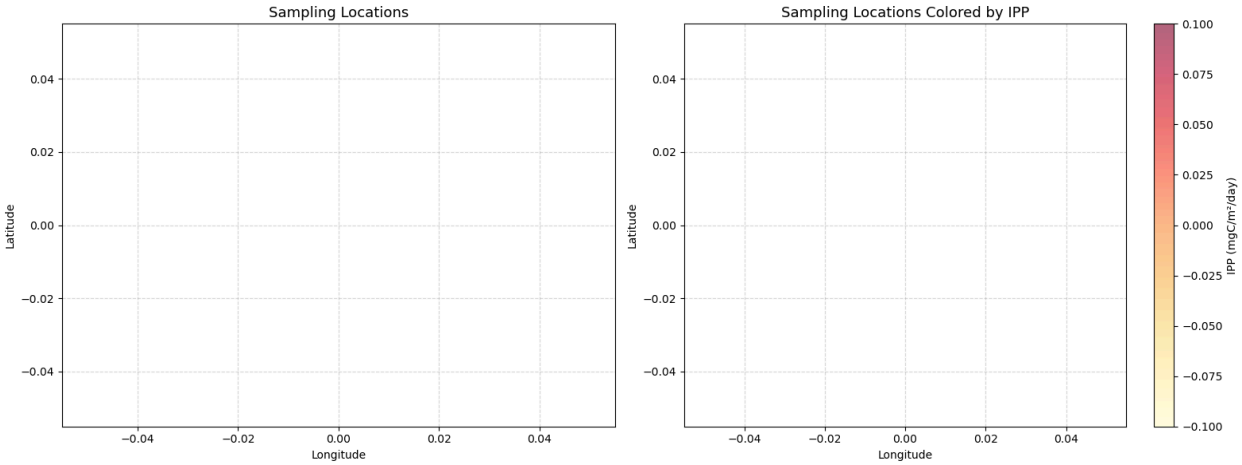


d. Dataset 4:
    i. EQPAC: Longitude data is also missing. Latitude values center around 12°N, consistent with equatorial

Pacific sampling. The equatorial Pacific is known for moderate productivity driven by equatorial upwelling.

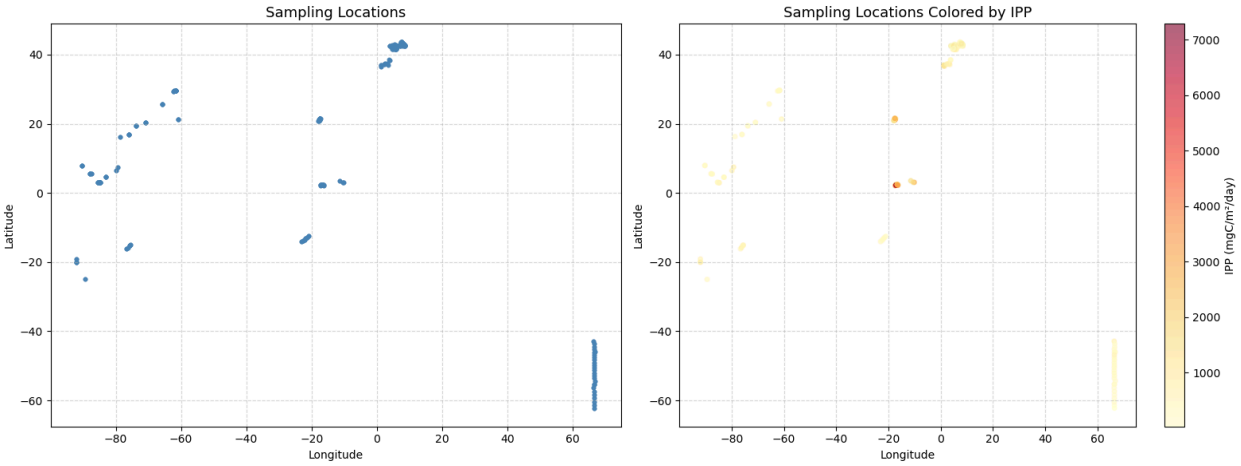


e. Dataset 5:
    i. BATS (Bermuda): Longitude missing. LAT ~31.7°N consistently, matching the Bermuda Atlantic Time-series Study location. Subtropical gyre generally has low to moderate productivity.

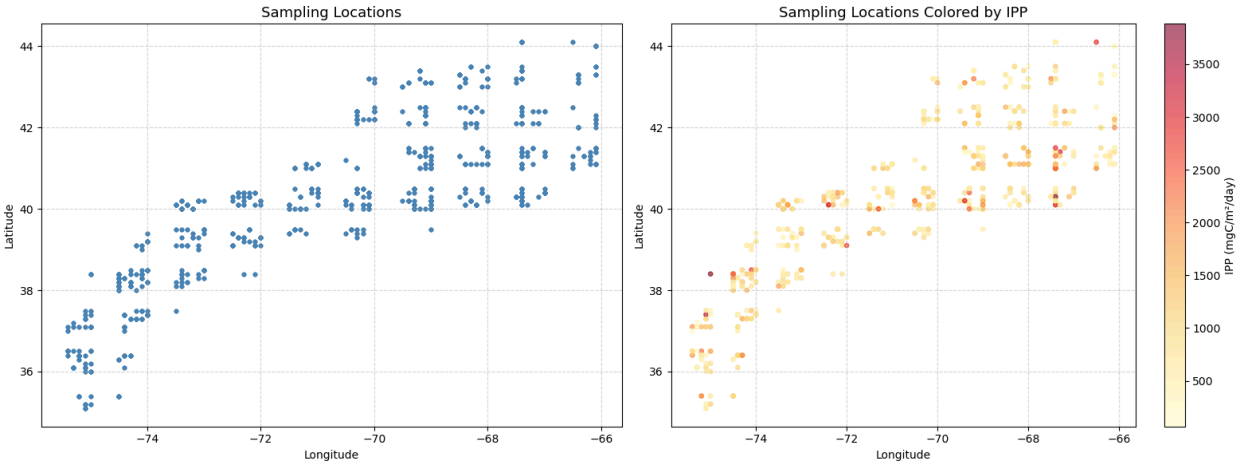


f. Dataset 6:
    i. Southern Ocean (Antiprod): Full LAT & LONG available. Sampling centered around LAT -45 to -46°S, LONG ~66°E, Indian Ocean sector of the Southern Ocean. This is one of the most

geographically distinct datasets, being the only one from the Southern Hemisphere.

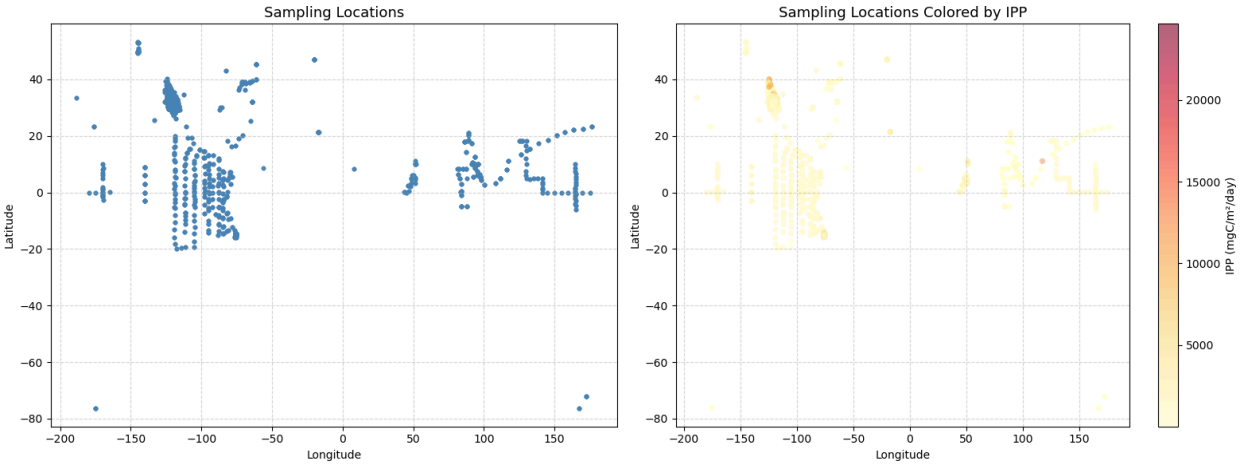


g. Dataset 7:
    i. marmap (Large dataset): 7,329 rows, but the LONG column is missing for most entries. LAT ~40°N suggests the US East Coast / mid-Atlantic region. This is the second-largest dataset.



h. Dataset 8:
    i. bbs (Largest dataset): 9,302 rows with full LAT/LONG available. Centered at LAT ~32°N, LONG -64°W,

Bermuda region. Data spans from 1958, making it the historically oldest dataset.



5. Three Issues that came up when processing this data:
    a. **Missing Longitude:** Files 3, 4, 5, and 7 have NaN values in the LONG column. This limits full geographic visualization for these datasets.
    b. **IPP only at surface row:** IPP is recorded only for the first depth measurement of each station (the integrated value for the whole water column). All sub-surface depth rows have NaN for IPP.
    c. **Small dataset size (File 2):** With only 71 rows and 4–5 unique stations, File 2 (Eumeli) has limited standalone modeling value. It will contribute more meaningfully when combined with other datasets.

6. Metadata:

Dataset	Cruise(s)	Time Period	Region	Rows
File 1	Albatros, Argus, Atlantis, Delaware, Eastward, Henlopen, Kelez, Knorr, NEWP, SEEP, and others	1975–1994	US East Coast + Arctic	1,165
File 2	Eumeli	1992	Tropical Atlantic (W. Africa coast)	71
File 3	HOTS	1988–1989	North Pacific subtropical (Hawaii)	249
File 4	EQPAC	1992	Equatorial Pacific	447
File 5	BATS	1988–1989	North Atlantic subtropical (Bermuda)	254
File 6	Antiprod + others	1977 onwards	Southern Ocean	1,741
File 7	marmap + others	1977 onwards	US East Coast + broader Atlantic	7,329
File 8	bbs + others	1958 onwards	North Atlantic / Bermuda region	9,302